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Strategies for predictive digital soil mapping by geophysical, remote sensing and machine learning approaches

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ABSTRACT

Pedology, the study of pedogenesis, includes soil classification and mapping. Digital soil mapping (DSM) has evolved from traditional methods to creating comprehensive spatial soil information systems. This advancement is achieved by integrating field and laboratory data with environmental covariates and incorporating new geotechnologies such as geophysical techniques and remote sensing data, alongside machine learning approaches. This integration in DSM provides novel insights into soil survey and mapping, offering detailed information on soil variability both vertically and laterally. It also raises new research questions that traditional pedology may not have addressed. In this study, we proposed and compared three strategies for DSM in Brazil, creating predictive pedological mapping. These strategies integrate data from three geophysical sensors, remote sensing data, relief, and lithology as input in a machine learning approach testing five algorithms. The four proposed strategies were: *i)* the combined use of geophysical variables and remote sensing data (G + RS + DEM); *ii)* the use of remote sensing data only (RS + DEM); *iii)* the use of geophysical variables only (G + DEM) and; iv) relief data (DEM). Lithology and relief were used as common input data in the predictive pedological mapping modeling process for all four strategies. We conducted a statistical analysis to evaluate the models' performance employing the Kruskal-Walli's test, the F1-score, Kappa, Accuracy, Sensitivity, and Specificity. Additionally, the best strategy was chosen based on the Kruskal-Walli's test and Overall Agreement and Disagreement statistical validation method, utilizing the reference map generated by an expert pedologist. Results revealed that the Random Forest algorithm presented the best performance for modeling predictive pedological mapping in all proposed strategies. Among the predictor variables, the Synthetic Soil Image (a synthetic multi-temporal soil image created by selecting and integrating bare soil observations from satellite data to capture key soil properties for mapping and analysis), relief, and geophysical data had the most significant contributions. While variables associated with remote sensing displayed stronger correlations with surface pedological attributes, geophysical variables demonstrated stronger associations with subsurface pedological attributes and diagnostic horizons. The most effective strategies for predictive digital pedological mapping were the G + RS, while the least effective was DEM. The individual performances of G and RS were comparable. The final predictive digital pedological map had a strong correlation with the traditional one, considering the Agreement/Disagreement validation method. The most significant prediction errors occurred in the transitional zones between pedological and lithological classes. Within the predicted classes, the most substantial errors were observed in classes exhibiting similar morphological attributes such as texture, color, and oxic mineralogy.

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1. Introduction

Information about the set of attributes that different types of soil possess is essential for various practical purposes related to sustainable land use and management. Such information is obtained through the classification and delimitation of the spatial distribution of soils, accomplished through survey work and pedological mapping (IUSS Working Group WRB, 2015). Inaccurate soil classification and mapping can lead to inappropriate soil use and management, resulting in harmful impacts on the environment and agriculture, forestry, and zootechnical production systems (Duniway et al., 2010).

Pedological mapping involves identifying and classifying soil types and their spatial distribution within a given geographic area (Rodrigo-Comino et al., 2018). Soil is defined as product of the interaction among soil formation factors identified by Dokuchaev (1967) and later formalized by Jenny (1941) as climate, organisms, relief, parent material, and time. Consequently, soil types vary spatially and temporally because their morphological attributes result from anisotropic geological, morphogenetic, and pedogenetic processes (Breemen and Buurman, 2003).

Two main approaches to soil mapping have been developed to document the spatial variation of soil resources worldwide. The first, known as conventional soil mapping systems, predominated between 1930 and the early 2000s. It relied on the interpretation of aerial photos, topographic maps, vegetation, and field observations, to create polygonal class maps of soil, assuming discrete boundaries between them (Hudson, 1992; Minasny and McBratney, 2016). The second approach, digital soil mapping (DSM), has emerged in the early 2000's following great advances in geotechnologies and computational capability (McBratney et al., 2003a). DSM accepts variability within soil classes and produces contour maps that reflect gradual transition zones. As a quantitative approach, DSM extends the classical soil formation factor equation to include new factors, providing a range of values for soil physical-chemical, mineralogical, and biological attributes using advanced statistical theories (Lagacherie et al., 2006; McBratney et al., 2003b).

Along with computational improvement, remote sensing (RS) and geophysical data have been available for soil researches use for over 40 years in different spatial resolutions and formats (Wooster et al., 2021). These combined technologies are being successfully recently used in soil survey and mapping, as well as in the spatial modeling of soil attributes by Mello et al., (Mello et al., 2022a; Mello et al., 2023a, 2023a; Mello et al., 2023a). This is due to the interaction between soil and various forms of energy that can be detected (Stenberg et al., 2010). The Geospatial Soil Sensing System (GEOS3), that creates the Synthetic Soil Image (SYSI) developed by Demattê et al. (2018) is a RS technique that uses multi-temporal satellite image information to construct a single image representing bare soil. The SYSI can be successfully used to support soil surveys and digital soil mapping, as demonstrated by Rizzo et al. (2020). The Geophysical proximal sensors, such as gamma-ray spectrometry, magnetic susceptibility (κ), and apparent electrical conductivity (ECa), have recently found applications in soil science, supporting DSM and pedogenetic processes evaluation, mainly gamma-ray spectrometry, magnetic susceptibility and apparent electrical conductivity (Mello et al., 2021; Mello et al., 2020; Mello et al., 2022b).

RS and on-site geophysical techniques, combined with advancements in machine learning approaches, have significantly enhanced our capability to gather, analyze, and predict soil spatial information. Despite these advancements, correlating this new data with soil attributes and processes, as well as identifying the most suitable soil survey and mapping strategies, remains challenging. Then, improved pedologic models are required.

While sensor fusion has been widely applied in digital soil mapping, integrating remote sensing and proximal geophysical data has become increasingly relevant in recent years (Chatterjee et al., 2021; Grunwald et al., 2015; Mouazen and Shi, 2021; Pozzuto et al., 2025; Schmidinger et al., 2024; Wang et al., 2022; Xue et al., 2023). Such studies have demonstrated the potential of combining multiple data sources to improve soil property prediction and mapping accuracy. However, few studies have rigorously evaluated the comparative advantages of integrating RS and geophysical datasets within machine learning frameworks, particularly in complex and heterogeneous landscapes. Our study addresses this gap by applying a multi-sensor fusion approach to improve digital pedological mapping, highlighting both methodological innovation and practical relevance for soil management and conservation. Therefore, our hypothesis is that combining multiple geophysical data with RS technology, which are correlated with soil subsurface and surface attributes, respectively, can improve digital pedological mapping using a machine learning approach.

Despite significant advancements in RS, geophysical techniques, and machine learning for soil spatial data analysis and prediction, there are persistent challenges in effectively correlating these datasets with soil attributes and processes. Current soil survey and mapping strategies often fail to fully leverage the synergistic potential of combining surface (RS) and subsurface (geophysical) data. Additionally, uncertainties in predictive pedological models remain high, highlighting the need for innovative approaches that integrate diverse datasets while improving model accuracy, validation, and reliability. Existing studies have yet to rigorously evaluate the comparative advantages of combining multiple data sources (e.g., RS, geophysical, and morphometric data) within machine learning frameworks, particularly in complex and heterogeneous landscapes like Brazil.

Recent advances in data fusion approaches have demonstrated the potential of combining proximal and remote sensing datasets to improve digital soil mapping accuracy and interpretability (Lin et al., 2024; Meng et al., 2022; Mouazen and Shi, 2021; Peng et al., 2024; Wadoux et al., 2020). These studies emphasize that integrating complementary data sources, particularly those capturing surface and subsurface soil properties can enhance the spatial prediction of soil attributes and classification reliability. However, most existing applications focus on soil properties such as texture, SOC, phosphorus, pH and other soil attributes related to soil fertility (Broeg et al., 2024; Mello et al., 2025; De Ros et al., 2025; Escorcia-Gutierrez et al., 2022; Gibson et al., 2021; Lilburne et al., 2024; Mgohele et al., 2024; Rosas et al., 2024) rather than soil classes, and few have systematically evaluated trade-offs between geophysical sensor types combinations and data acquisition costs. By situating our work within this context, we highlight the novelty of our comparative framework, which assesses both methodological performance and practical implications of geophysical sensor integration for predictive pedological mapping.

In this study, we developed and evaluated four distinct strategies for mapping soil types across Brazil using machine learning, testing multiple algorithms for their effectiveness. The first strategy combined data from geophysical field sensors and remote sensing (G + RS + DEM), the second strategy used only remote sensing data (RS + DEM), the third relied solely on geophysical variables (G + DEM), and the fourth strategy used only morphometric relief variables (DEM).

The main novelty of this study lies not in the isolated or combined use of multiple datasets, but in the systematic and comparative evaluation of feature-level multi-sensor integration strategies for predictive digital pedological mapping. Specifically, we assess different soil survey-mapping strategies by combining complementary geophysical techniques, rarely evaluated together, with synthetic soil images acting as remote sensing proxies, relief data within machine learning frameworks. This design allows a direct assessment of the marginal contribution, redundancy, and trade-offs associated with integrating surface and subsurface information, while also enabling the robustness of predicted pedological units to be evaluated using both empirical accuracy metrics and agreement-disagreement validation approaches.

2. Materials and methods

2.1. Study site and conventional soil-lithological survey and mapping

The investigation was conducted within a rural expanse spanning 165 ha situated in the Piracicaba municipality, São Paulo state, Brazil, pinpointed at coordinates WGS 1984–22.792333° S, −47.571245° W (Fig. 1). The area has been historically cultivated with sugar cane for the past 10 years, with plowing and harrowing management conducted for each harvest, along with the application of limestone, vinasse, and fertilizers relevant to the crop. The terrain, ranging in elevation from 543 to 644 m, encompasses steep gradients fostering soil diversity. Its geological composition comprises igneous rock (basalts), sedimentary rocks (siltites and sandstones), basalt-siltites (metamorphosed siltite), and fluvial deposits (Fig. 2). Notable soil classifications within the study site, as per World Reference Base for Soil Resources (FAO, 2014), include Haplic Cambisol (CX), Haplic Plinthosol (FX), Dystric Gleysol (GX), Rhodic Ferralsol (LV), Rhodic Nitisol (NV), Xanthic Acrisol (PA), Rhodic Acrisol (PV), Haplic Acrisol (PVA), Lithic Leptosol (RL) (Fig. 2). In the study area, no association-type mapping units were identified; instead, it was possible to differentiate all taxonomic classes into simple mapping units.

The detailed pedological survey and mapping, executed at a scale of 1:10,000, were conducted in the field by a proficient pedologist. Twenty-seven soil profiles were morphologically described, and samples were collected in each soil horizon for laboratory physical and chemical analysis. The soils were classified according to the Brazilian Soil Classification System (dos Santos et al., 2018) and then reclassified according to the World Reference Base for Soil Resources system (FAO, 2014). In addition, 183 soil samples were collected for physicochemical analysis using augering. Also, the pedologist used a synthesis of RS, on-site investigation, and soil data, yielding detailed maps with a resolution of 30 m. The soil mapping process was carried out according to IBGE (2007) and Soil Survey Staff (2017) in the following sequence: objectives were defined, and pre-existing information about the area, including historical data, base maps, and prior soil maps, was gathered. A general field survey was conducted to assess the terrain, and sampling points were marked on the base map, prioritizing transitions and topographic sequences between soils. Soil samples were collected at various depths—typically from diagnostic horizons in soil profiles and at depths of 0–20 cm, 40–60 cm, and 80–100 cm at augering sampling points. Characteristics such as color, texture, and gravel content were recorded. The samples were then sent for chemical and granulometric analyses. Preliminary soil units were identified in the office based on field observations and contour lines. To refine the soil classification, trenches were excavated and analyzed in homogeneous areas. Finally, the soils were classified, inconsistencies were resolved, and the final soil map was produced. A descriptive report detailing the methodologies and observations was also prepared. The selection of this area was primarily motivated by its intricate landscape, anticipating a rich assortment of parent materials and soil types, facilitating effective sensor exploration. While a lithological map delineated by Bonfatti et al. (2020) at a scale of 1:100,000 was available, we opted for a customized adaptation at 1:10,000 meticulously crafted by an expert pedologist through extensive field and laboratory work.

The climatic conditions adhere to a subtropical mesothermal classification according to the Köppen system, characterized by an average annual precipitation of 1200 mm and mean temperatures of 18 °C during winter and 22 °C during summer, as reported by Alvares et al. (2013).

2.2. Geophysical surveys

The geophysical survey was undertaken by using combined geophysical techniques: gamma-ray spectrometry, soil magnetic susceptibility and soil apparent electrical conductivity. Gamma-ray spectrometry involves measuring natural gamma radiation emitted by potassium (K^{40}), uranium (U^{238}) and thorium (Th^{232}) quantifying also the total radionuclides elements content, and total emissions from soil elements (Dickson and Scott, 1997). Soil magnetic susceptibility (κ)

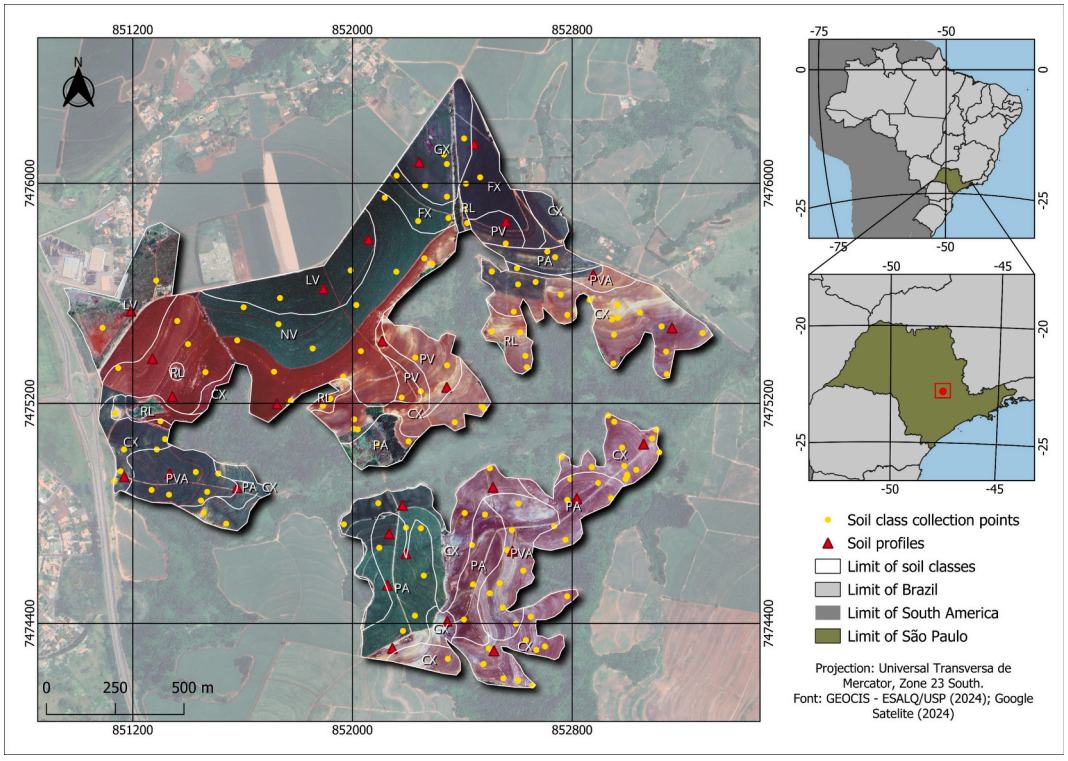


Fig. 1. Study site, soil types, and collected sampling points for soil physico-chemical analyses and geophysical sensor readings. CX: Haplic Cambisol; FX: Haplic Plinthosol; GX: Dystric Gleysol; LV: Rhodic Ferralsol; NV: Rhodic Nitisol; PA: Xanthic Acrisol; PV: Rhodic Acrisol; PVA: Haplic Acrisol; RL: Lithic Leptosol.

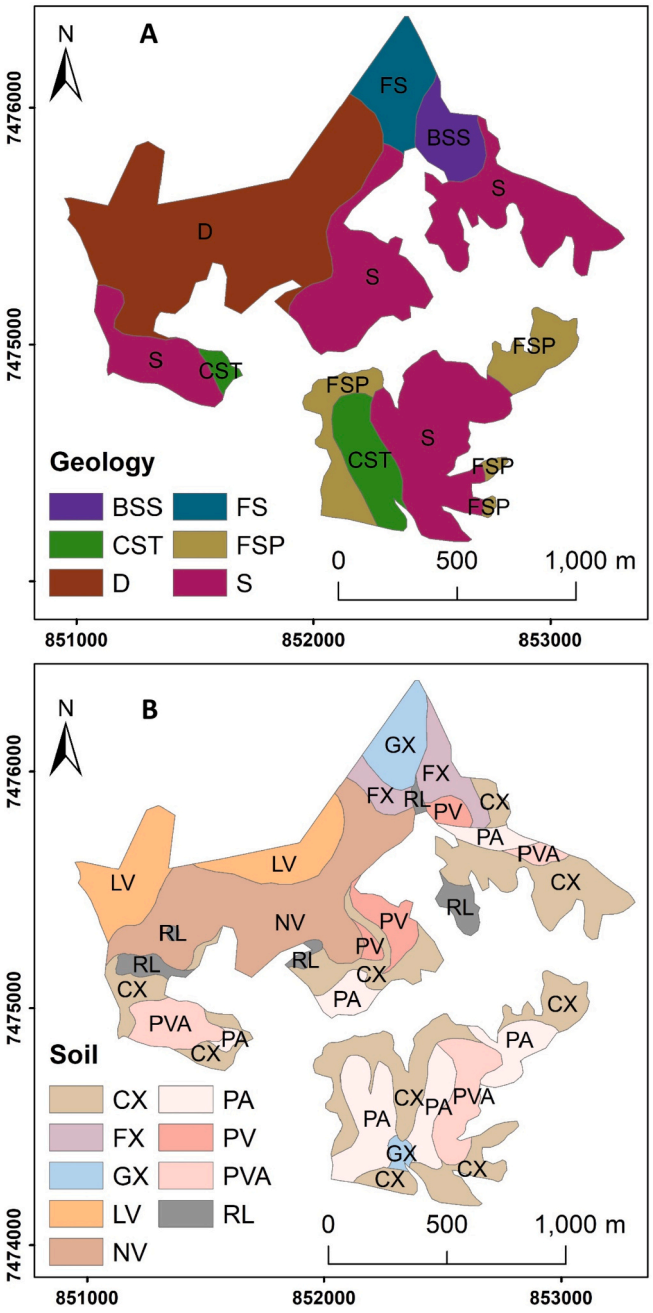


Fig. 2. A) Lithology: BSS: basalt-siltite system; CST: sandstone-tubarão; D: Diabase; FS: Fluvial sediments; FSP: sandstone-pirambóia; S: siltite; B) Soil classes: CX: Haplic Cambisol; FX: Haplic Plinthosol; GX: Dystric Gleysol; LV: Rhodic Ferralsol; NV: Rhodic Nitisol; PA: Xanthic Acrisol; PV: Rhodic Acrisol; PVA: Haplic Acrisol; RL: Lithic Leptosol.

measures the magnetization capacity of soil particles due to ferrimagnetic minerals (Rochette et al., 1992). Soil apparent electrical conductivity (ECa) indicates the soil's ability to conduct electricity (Corwin and Lesch, 2003).

The geophysical survey data and soil samples were gathered at 183 designated sampling points and at soil surface at the 27 soil profiles used for soil classification (Fig. 1) employing the three proximal geophysical sensors: the gamma-ray spectrometer RS-230, the magnetic susceptibility meter KT-10 Terraplus, and the conductivimeter EM38-Geonics. These sensors allowed the collection of soil gamma-ray/radiometric data, soil κ, and soil ECa, respectively (Table 1). In addition, all data acquired from the geophysical sensors were integrated with the soil analytical data collected at the same locations where the sensor measurements were performed.

2.2.1. Gamma-ray survey

Soil gamma-ray spectrometric characterization was conducted using a proximal-surface gamma-ray spectrometer (GM), the RS 230 model by Radiation Solution Inc. (Ontario, Canada). This instrument measures eTh and eU in parts per million (ppm), while K^{40}, due to its relative abundance in the soil, is reported in percentage. Standard units for radionuclide concentrations are mg kg^{-1} for eU and eTh, and percentage for K^{40}. The GM spectrometer detects gamma-ray emissions to a depth of 30–60 cm, with detection depth varying due to factors like soil bulk density and moisture (Beamish, 2015; Taylor et al., 2002; Wilford, 1997).

Initially, the GM was calibrated by placing it on the ground surface and leaving it active for five minutes, allowing the readings of eU, eTh, and K^{40} levels to stabilize (Solutions, 2009). Radionuclide measurements were taken using the "assay mode" for maximum precision, with the GM positioned at the soil surface for two minutes at each of the 183 sampling points. Geographic coordinates for each point were recorded with a GPS integrated into the GM (GPS – Radiation Solution Inc., Ontario, Canada) with a precision of 1 m.

2.2.2. Magnetic susceptibility (κ) survey

For soil κ analysis, measurements were taken at all 183 sampling points using a geophysical susceptibility meter (KT10 – Terraplus). This device measures κ to a depth of 2 cm below the surface with a precision of 10^{-6} SI units, expressed in m^3 kg^{-1}. To ensure accuracy, the sensor was calibrated by first determining the frequency of the outdoor oscillator. Readings were then conducted in three sequential steps: (1) recording the frequency and amplitude of the oscillator in free air; (2) placing the coil directly on the soil surface to measure frequency and amplitude on the sample; and (3) repeating the initial free-air measurement and displaying results. Details on these procedures can be found in (Sales, 2021). Measurements were taken in scanner mode, optimizing the geometric correlation for direct κ readings to achieve rapid and accurate results. Three readings were conducted in a triangular pattern around each sampling point, and the mean κ value was used in all analyses to minimize noise.

2.2.3. Apparent electrical conductivity (ECa) survey

We conducted the soil ECa measurements using the Geonics EM38 conductivity meter (Geonics Ltd., Mississauga, Ontario, Canada) (McNeill, 1986). The EM38 measures quad-phase conductivity (in m S m^{-1}) without requiring direct soil contact (Geonics, 2002). Calibration followed the protocol described by Heil and Schmidhalter (2019), Section 3.1.1, as ECa values depend on calibration settings, coil orientation, and coil separation (Heil and Schmidhalter, 2019). Further operational details on the EM38 can be found in Hendrickx et al. (2002). Following calibration, ECa readings were taken at each of the 183 sampling points using a vertical dipole orientation, which provides data from an effective depth of approximately 1.5 m. Field data were collected during the dry season on bare soil and at consistent intervals to minimize environmental influences. To prevent interference, all metal objects were kept at a distance from the EM38 during readings.

Table 1
Geophysical data used in pedological predictive modeling process (*input data*).

Geophysical data	Abbreviations
Equivalent uranium	eU
Equivalent thorium	eTh
Potassium 40	K^{40}
Soil apparent electric conductivity	ECa
Soil magnetic susceptibility	κ

2.3. Synthetic soil image (SYSI)

The SYSI is an approach that combines multi-temporal satellite images to create a single synthetic image representing soil characteristics over time. SYSI typically identifies bare soil through multi-temporal analysis, selecting images or pixels from periods when vegetation cover is minimal or absent. These bare soil observations are then integrated over time to build the synthetic SYSI image, ensuring that the soil signal is strong and representative of the underlying properties. The system involves acquiring satellite images from different periods to capture seasonal and temporal variations, processing these images to correct distortions and align them, and integrating the processed data into a comprehensive synthetic representation. Machine learning algorithms and statistical models are then applied to analyze the integrated data, allowing the identification and quantification of soil attributes such as texture, moisture, and mineral composition. This approach provides detailed and accurate information for digital soil mapping, precision agriculture, and sustainable land management (Demattê et al., 2018).

The Synthetic Soil Image (SYSI) was generated using the Geospatial Soil Sensing System (GEOS3) methodology, as described by Demattê et al. (2018). The SYSI was developed using the complete time series of images from the Landsat archive (USGS, 2019a, 2019b), spanning from 1982 to 2023. This image reflects the average bare soil reflectance and includes six spectral bands: Blue (band 1), Green (band 2), Red (band 3), Near-Infrared (NIR - band 4), Shortwave Infrared 1 (SWIR1 - band 5), and Shortwave Infrared 2 (SWIR2 - band 6).

Landsat images were selected due to their greater availability compared to Sentinel, offering more extensive temporal coverage, which was sufficient to capture a large number of exposed soil images. The approach involved processing Landsat satellite images captured during the dry seasons within the study area while excluding irrelevant features. The GEOS3 methodology focused on satellite imagery from dry seasons between 1982 and 2023, chosen for their reduced cloud cover, increased visibility of bare soil, and minimized moisture influence on the soil surface. Further details on the GEOS3 methodology can be found in Demattê et al. (2018) and Fongaro et al. (2018).

2.4. Digital elevation model and acquisition of terrain attributes

The digital elevation model (DEM) was generated using a 1:10,000-scale planialtimetric map as the foundational base, supplemented with 5-m contour lines, resulting in a DEM with a 5-m resolution and high accuracy. ESRI ArcGIS 10.4 software was utilized for geoprocessing of the data. The initial map was compiled by the Campinas Geographic Institute (IGC). Subsequently, 48 additional morphometric variables (Table 2) were extracted from the final DEM using the SAGA GIS by using SAGA tools (https://saga- gis.sourceforge.io/saga_tool_doc/6.2.0/ta_lighting.html; https://saga-gis.sourceforge.io/saga_tool_doc/6.2.0/ta_morphometry.html) (Brenning, 2008) and R software (R Core Team, 2023). These spectral and morphometric covariates (Table 2) were stacked with the SYSI bands to form the foundational dataset for modeling.

2.5. Predictive pedological mapping and modeling processes

The input data included the target variable (soil classes) and various predictor variables, such as geophysical data (eU, eTh, K^{40}, soil κ, and soil ECa), remote sensing data (SYSI index), as well as lithological and terrain morphometric covariates derived from the digital elevation model (Tables 1 and 2). Based on this data, four pedological mapping strategies were developed. For each of these four strategies, different sets of predictor variables were tested, resulting in the following combinations:

***i)* Geophysics + Remote Sensing (G + RS + DEM) strategy**, with input data: eU, eTh, K^{40}, soil κ, soil ECa, and SYSI;

Table 2
Terrain attributes generated from the digital elevation model.

Terrain attributes	Abbreviations	Brief description
Aspect	AS	Slope orientation
Convergence index	CI	Convergence/divergence index in relation to runoff
Cross sectional curvature	CSC	Measures the curvature perpendicular to the down slope direction
Diurnal anisotropic heating	DAH	Continuous measurement of exposure dependent energy
Flow line curvature	FLC	Represents the projection of a gradient line to a horizontal plane
General curvature	GC	The combination of both plan and profile curvatures
Hill	HI	Analytical hill shading
Hill index	HIINDEX	Analytical index hill shading
Longitudinal curvature	LC	Measures the curvature in the down slope direction
Maximal curvature	MAXC	Maximum curvature in local normal section
Mid-slope position	MSP	Represents the distance from the top to the valley, ranging from 0 to 1
Minimal curvature	MINC	Minimum curvature for local normal section
Morphometric Protection Index	MPI	Measure of exposure/protection of a point from the surrounding relief
Multiresolution index of ridge top flatness	MRRTF	Indicates flat positions in high altitude areas
Multiresolution index of valley bottom flatness	MRVBF	Indicates flat surfaces at bottom of valley
Normalized height	NH	Vertical distance between base and ridge of normalized slope
Plan curvature	PLANC	Described as the curvature of the hypothetical contour line passing through a specific cell
Profile curvature	PROC	Describes surface curvature in the direction of the steepest incline
Real surface area	RSA	Actual calculation of cell area
Slope	S	Represents local angular slope
Slope height	SH	Vertical distance between base and ridge of slope
Slope Index	SI	Represents a local angular slope index
Solrad Diffuse1	SolDiffuse1	Insolation Diffuse for the month of January
Solrad Diffuse2	SolDiffuse2	Insolation Diffuse for the month of July
Solrad dur 1	SolDur1	Insolation Duration for the month of January
Solrad dur 2	SolDur2	Insolation Duration for the month of July
Solrad Direct1	SolDiret1	Insolation Direct of month January
Solrad Direct2	SolDiret2	Insolation Direct of the month of July
Solrad Ration1	SolRation1	Ratio between direct Insolation and diffuse Insolation of the month of January
Solrad Ration2	SolRation2	Ratio between direct Insolation and diffuse Insolation of the month of July
Solrad Sunrise1	SolSunrise1	Mean sunrise time of month January
Solrad Sunrise2	SolSunrise2	Mean sunrise time of month July
Solrad Sunset1	SolSunset1	Mean sunset time of month January
Solrad Sunset2	SolSunset2	Mean sunset time of month July
Solrad total1	SolTotal**1**	Total Insolation for the month of January
Solrad total2	SolTotal**2**	Total Insolation for the month of July
Standardized height	STANH	Vertical distance between base and standardized slope index
Surface specific points	SSP	Indicates differences between specific surface shift points

(*continued on next page*)

Table 2 (*continued*)

Terrain attributes	Abbreviations	Brief description
Tangential curvature	TANC	Measured in the normal plane in a direction perpendicular to the gradient
Terrain ruggedness index	TRI	Quantitative index of topography heterogeneity
Terrain surface convexity	TSC	Ratio of the number of cells that have positive curvature to the number of all valid cells within a specified search radius
Terrain surface texture	TST	Splits surface texture into 8, 12, or 16 classes
Total curvature	TC	General measure of surface curvature
Topographic position index	TPI	Difference between a point elevation with surrounding elevation
Valley depth	VD	Calculation of vertical distance at drainage base level
Valley	VA	Calculation fuzzy valley using the Top Hat approach
Valley Index	VA	Calculation fuzzy valley index using the Top Hat approach
Vector ruggedness measure	VRM	Measures the variation in terrain roughness
Topographic wetness index	TWI	Describes the tendency of each cell to accumulate water as a function of relief

ii) **Geophysics (G + DEM) strategy**, with input data: eU, eTh, K^{40}, soil κ, and soil ECa;

iii) **Remote Sensing (RS + DEM) strategy**, with input data: SYSI data.

iv) **Morphometric relief variable (DEM) strategy**: all relief data presented in Table 2.

In all four strategies, core data, including additional relief attributes and lithology, were integrated into the predictive pedological model, with an existing soil map serving as the reference for validation using the Agreement/Disagreement statistical method. Finally, these strategies were compared to a baseline method that used only the digital elevation model (DEM) for relief data in mapping soil classes. Each strategy was assessed for accuracy and reliability, considering model performance and statistical measures of Agreement and Disagreement, to identify the most effective approach for soil mapping in Brazil.

In addition to the geophysical (G + DEM) and remote sensing (RS + DEM) variables specific to each strategy, lithological data were included as predictors across all three evaluated strategies. Lithology was the only categorical predictor variable used in this analysis.

We compared the three tested strategies with the individual use of relief morphometric variables in predictive modeling of pedological mapping. However, since this approach is already commonly used, we did not consider it a new strategy in this work.

Five algorithms from different families were tested in a machine learning approach for predictive modeling of pedological classes across the three strategies: Model Averaged Neural Network (avNNet – Feed-Forward Neural Networks), C5.0 (C5.0 – Decision Trees), k-Nearest Neighbors (kknn - Nearest Neighbors), Random Forest (RF - Decision Tree), and Support Vector Machines with Radial Basis Function Kernel (SVM - Support Vector Machine).

The five machine learning algorithms evaluated in this study were selected to represent different algorithmic families, operating through distinct learning mechanisms, including tree-based, ensemble, instance-based, and kernel-based methods. This selection strategy enables a systematic comparison of the predictive performance of different learning paradigms under the same set of covariates and experimental conditions, thereby providing a more comprehensive assessment of model behavior and robustness for the proposed application.

Averaged Neural Network (AvNNet).

Neural networks are computational models designed to emulate the human brain by identifying patterns and relationships within data. They are adaptable to varying inputs, producing optimized outputs without predefined patterns. Neural networks are commonly used in artificial intelligence and automated systems. The Averaged Neural Network (AvNNet) approach involves training multiple neural networks on the same dataset and averaging their predictions to improve robustness. This approach is analogous to the Random Forest algorithm, which aggregates predictions from decision trees (Kaur et al., 2022). Each neural network in the AvNNet method has three interconnected layers of neurons. Variability in predictions arises due to factors such as random seed initialization or bootstrap sampling during training. For classification tasks, the final prediction is determined by averaging class probabilities across all networks.

Kernel K-Nearest Neighbors **(KKNN).**

The weighted Kernel K-Nearest Neighbors (KKNN) algorithm is an advanced version of the traditional KNN method. It offers greater flexibility by incorporating locally weighted regression principles, similar to Loess regression, which fits smooth curves through scatterplots using local weighting (Schliep, 2004). In KKNN, weights are assigned to neighbors based on their proximity to the target observation, with closer neighbors receiving higher weights. Unlike standard KNN, which assumes equal influence from all neighbors, KKNN transforms distances into similarity measures that act as weights for predictions. This weighting enhances the model's predictive performance, particularly in non-linear data distributions.

C5.0.

The C5.0 algorithm is a supervised learning model that uses labeled data to construct predictive decision trees (Klein et al., 2012). It generates interpretable if-then rules, making the results easy to understand and apply (Siknun and Sitanggang, 2016). The C5.0 model is built recursively (Quinlan, 1993) using training data and conditioning factors. In its implementation, the Gain Ratio (GR) index is used to determine splits that maximize data homogeneity (Quinlan, 2014). This index also helps mitigate overfitting, ensuring the model generalizes well to unseen data. The tree-based C5.0 model constructs a single, highly accurate decision tree by optimizing splits at each node (Kotsiantis et al., 2007).

Random Forest (RF).

Random Forest (RF) is a machine learning algorithm developed by Breiman (2001) that is based on an ensemble of decision trees. It is widely used for classification and regression tasks. RF builds multiple decision trees during training and combines their outputs to produce a final prediction, using the average for regression tasks and the majority vote (mode) for classification (Breiman, 2002). Each tree is constructed using the bootstrap sampling method, where random subsets of the training data are created, with approximately one-third of the data reserved as out-of-bag (OOB) samples. The nodes of each tree are constructed from unique bootstrap samples of the original dataset. The OOB samples are used to estimate the training error and to assess the relative importance of predictors.

Support Vector Machine (SVM).

SVM is a machine learning method based on hyperplane theory, used for classification and regression. In regression, it solves convex optimization problems using quadratic programming and the ε-insensitive loss function (Vapnik, 1968). SVMs employ kernel functions—commonly linear, polynomial, or radial basis function (RBF) to map input data into higher-dimensional spaces for optimized modeling. RBF-based models optimize two key parameters: “sigma,” controlling nonlinearity, and “cost,” balancing model complexity and training error. In this study, SVM with RBF kernel (SVMRadialSigma) was applied, with “sigma” and “cost” optimized for improved performance.

The general process of predictive modeling of pedological classes, including input variables, evaluated mapping strategies, and statistical analyses, is demonstrated in the flowchart in Fig. 3.

The input data from the sets of predictor variables were subjected to

covariate removal/selection techniques, model optimization, and spatialization of soil classes for the study area.

The selection/removal of covariates was carried out to reduce their final number in the model, aiming for a more parsimonious result and lowering the computational cost of optimizing the final model (Mello et al., 2023b; de Paes et al., 2022). This removal process was conducted in three phases: removal by variance, removal by correlation, and removal by importance. The first two phases were performed on the full set of covariates, while the third step was applied only to the training set.

The first phase involved removing covariates with variance close to or equal to zero, using the '*nearZeroVar*' function from the Caret package (Kuhn, 2022). This function is described by (Mello et al., 2023b; de Paes et al., 2022). The second phase focused on removing covariates by correlation, using the '*findCorrelation*' function with *Spearman's* rank correlation coefficient and a 95% correlation threshold (Kuhn, 2022). The function evaluates predictor variables in pairs, identifying those with values above the defined linear threshold and eliminating the variable with the highest absolute sum of correlations with all other

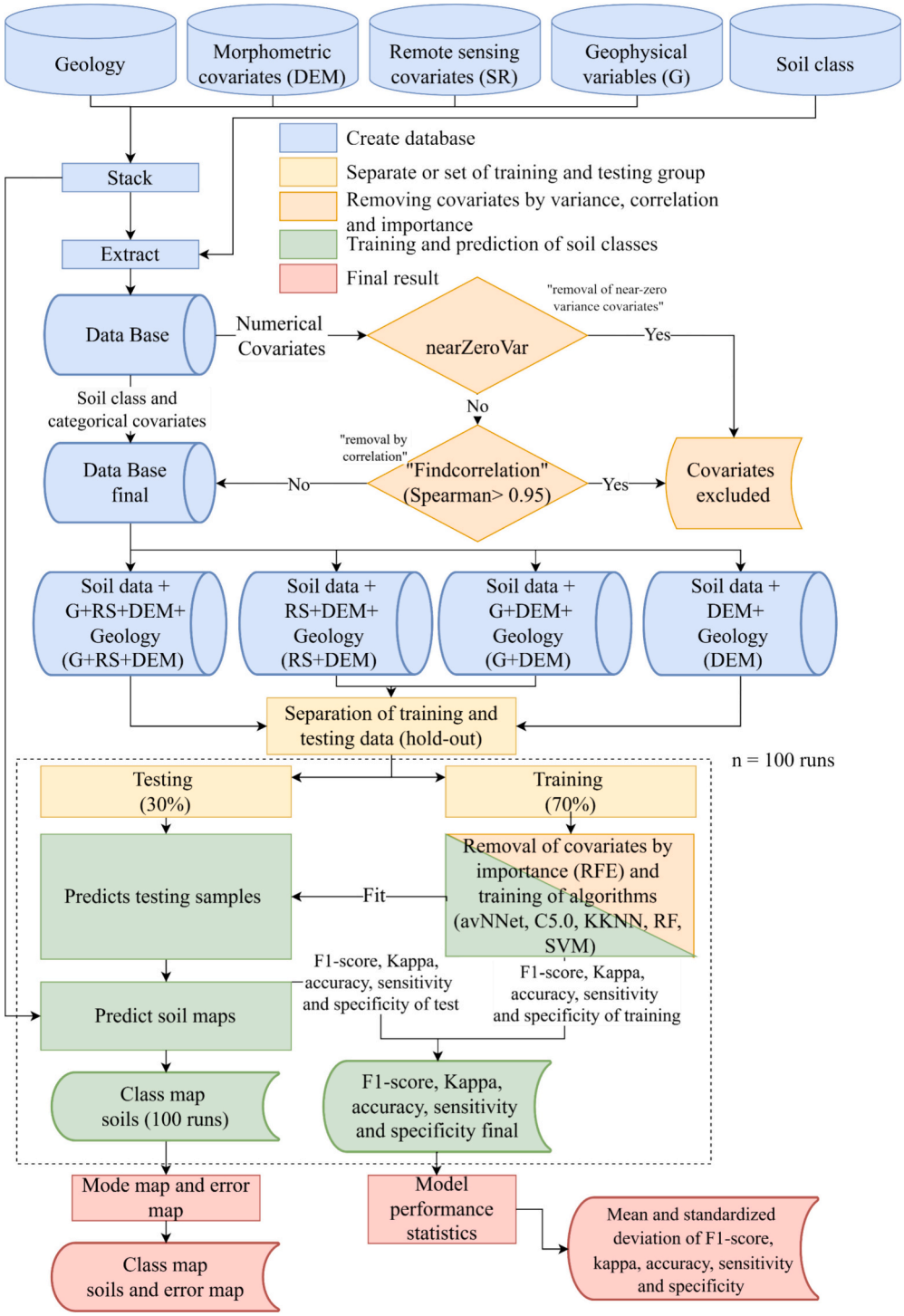


Fig. 3. Methodological flowchart showing the sequence of methodologies applied for predictive soil mapping strategies: *i)* the combined use of geophysical variables and remote sensing data (G + RS + DEM); *ii)* the use of remote sensing data only (RS + DEM); *iii)* the use of geophysical variables only (G + DEM) and; *iv)* relief data (DEM).

predictor variables. The third phase involved combining the results from this phase with categorical covariate (geology).

The dataset was subsequently divided into training (70%) and test (30%) sets using the hold-out method. This method ensures that the sample distribution in the two groups remains equivalent to the distribution in the original dataset. The separation process was conducted using the *createDataPartition* function from the "caret" package in the R software. The training set was used in the third removal phase (importance). At this stage, the Recursive Feature Elimination (RFE) function was applied to each tested algorithm. Each algorithm generates a unique set of covariates based on their predictive importance. We tested the most common algorithms from five different families: Model Averaged Neural Network (avNNet-Neural Network), C5.0 (C5.0-Decision Trees – tree and rules), k-Nearest Neighbors (kknn-Nearest Neighbors), Random Forest (RF-Decision Tree) and Support Vector Machines with Radial Basis Function Kernel (SVM-Support Vector Machine). The RFE function was run with 19 sets of covariates, from 5 to 20, 25 and the full set of covariates, which varies depending on the modeling strategy. The choice of the best set of covariates was carried out using cross validation with 10 folds, one repetition ("CV") and the highest F1-score value for the sets of participants. The RFE method is described by de Souza et al. (2022); Gomes et al. (2019). The selected (optimal) set of covariates was used to optimize the final algorithms.

The final model of each algorithm was optimized using the ideal covariate sets obtained from the importance removal phase for each algorithm. This optimization was performed with repeated cross-validation (*repeatedCV*) using 10 folds and three repeats. Five internal values of each algorithm (*tuneLength/hyperparameters*) were tested to optimize the model, using the F1-score as the performance parameter.

Hyperparameters are internal control variables within machine learning algorithms that directly influence the training process and, consequently, the performance of the final model. Each algorithm has a specific set of hyperparameters, which may assume numerical or categorical values within a predefined range.

The selection and tuning of hyperparameters were performed using the train() function from the **caret** package in R environment, which automates the tuning process. This function systematically explores the hyperparameter search space by evaluating multiple combinations of values evenly distributed within the predefined range. For each combination, the model is trained and validated, typically through *k*-fold cross-validation to assess its predictive performance according to a chosen metric (e.g., F1-Score, Kappa, or accuracy). The configuration that yields the best average performance across folds is then selected as the optimal set of hyperparameters. This procedure ensures a reproducible and objective tuning process, minimizing user bias in parameter selection and improving model generalization. All hyperparameter options and tuning ranges followed the guidelines described in Chapter 6 of the Caret manual, 'Models described', available at caret models (https://topepo.github.io/caret/train-models-by-tag.html).

After optimization, the final soil class map for the study area was generated, and the performance parameters were calculated based on each optimized model for each algorithm used in the study. In addition to the F1-score, the performance parameters Kappa, Accuracy, Sensitivity, and Specificity were evaluated.

The importance of predictor variables for each algorithm was determined using the internal methods that each algorithm employs for this calculation. For algorithms that do not have a native method for calculating importance (such as SVM), the procedure described in Chapter 15 of the caret manual was used, available at caret models (https://topepo.github.io/caret/variable-importance.html).

2.6. Final results, statistical analyses and model performance evaluation

The processes of selection and removal of covariates by importance, as well as training, mapping, and evaluation of the optimized model, were repeated 100 times, using different training and testing sets in each run. This procedure aims to increase the accuracy and reliability of the final result (Gomes et al., 2019). The performance parameters were calculated as the average of these 100 runs for each algorithm and pedological modeling strategy. The final maps were generated using the mode of these 100 iterations for each pixel in the map. Additionally, the prediction error was calculated, indicating how many times the algorithm failed to predict the most frequent class in each pixel of the map (mode).

To determine the best algorithm and sampling strategy, we applied the Kruskal-Wallis test and Dunn's post-hoc test, with a significance level of 5%, using the F1-score as the performance parameter (EQ. 1). The nonparametric Kruskal–Wallis test (with a 5% significance level) was performed to select the best evaluated strategies. In addition, to verify statistical differences among the algorithms tested within each strategy, Dunn's post hoc test (with a 5% significance level) was conducted based on the F1 score, kappa, accuracy, sensitivity, and specificity metrics. We also evaluated the performance parameters Kappa, Accuracy (EQ. 2), Sensitivity (EQ. 3), and, Specificity (EQ. 4) and Kappa (EQ. 5).

The F1-score test serves as a performance parameter to evaluate classification models in machine learning. It calculates the harmonic average of precision and recall, making it suitable for assessing datasets with imbalanced classes (Sasaki, 2007). This parameter effectively evaluates binary classification systems containing only 'positive' or 'negative' values. By factoring in false positives and false negatives, the F1-score offers a comprehensive assessment of the model's performance.

$$\mathrm{F-1\ Score} = 2^{*}(Precision.Recall)/(Precision + Recall) \tag{1}$$

$$\mathrm{F-1\ Score} = (T.P)\Big/\left[(T.P) + \frac{1}{2}(F.P + F.N)\right]$$

where:

T.P = True positive.

F·P = False positive.

F·N = False negative.

The accuracy metric assesses the overall performance of the final prediction model by tallying the number of correct predictions. It indicates the probability that the classified classes correspond to the actual data, with values ranging from 0 to 1.

$$\mathrm{Accuracy} = \frac{\sum_{i=1}^{n} x_i}{n}\ x\ 100 \tag{2}$$

where:

x_i = number of correctly classified samples for class ii;

k = number of classes;

n = total number of samples.

Mean sensitivity refers to a model's ability to accurately predict the correct values of a class, while average specificity indicates the model's ability to correctly identify values that do not belong to that class.

$$Sensitivity = \frac{\sum_{1}^{n}\left(\frac{TP}{TP+FN}\right)}{n} \tag{3}$$

$$Specificity = \frac{\sum_{1}^{n}\left(\frac{TN}{TN+FP}\right)}{n} \tag{4}$$

where:

TP = number of true positives.

FP = false positives.

TP = false negatives.

n = number of existing classes.

The Kappa index provides a numerical measure that quantifies the level of agreement between prediction results and a reference value. Derived from the confusion matrix (Morales et al., 2018) (Eq. 5), the

Kappa value ranges from 0 (indicating no agreement) to 1 (indicating almost perfect agreement) (Landis and Koch, 1977).

$$\text{Kappa} = \frac{n\sum_{i=1}^{c} n_{ii} - \sum_{i=1}^{c} n_{i+} + n_{+i}}{n^2 - \sum_{i=1}^{c} n_i + n_{+i}} \quad (5)$$

where:

K = Kappa estimate;

n_{ii} = the value in row i and column i (observed agreement);

n_{i+} = sum of row i, and n + i is the sum of column $_i$ of the confusion matrix (product of the marginals, being the expected agreement);

n = total number of samples;

C = total number of classes.

The algorithms and sampling strategies that presented the best statistical results compared to others were chosen. The best results for the four strategies were evaluated by Overall Agreement and Overall Disagreement using the map generated by the expert as the reference map (Pontius Jr and Millones, 2011; Silva et al., 2023). All processing was carried out in R 4.2.0 (R Core Team, 2023).

The percentage of success and error was calculated based on the Agreement/Disagreement statistical validation test, comparing the maps predicted by the RF model with those observed through traditional pedological mapping for each predictive strategy. This evaluation involved assessing the differences between the model-predicted maps and the reference map created by an experienced pedologist. The comparison was performed pixel by pixel, where areas with similar results between the two maps were considered correct predictions, and areas

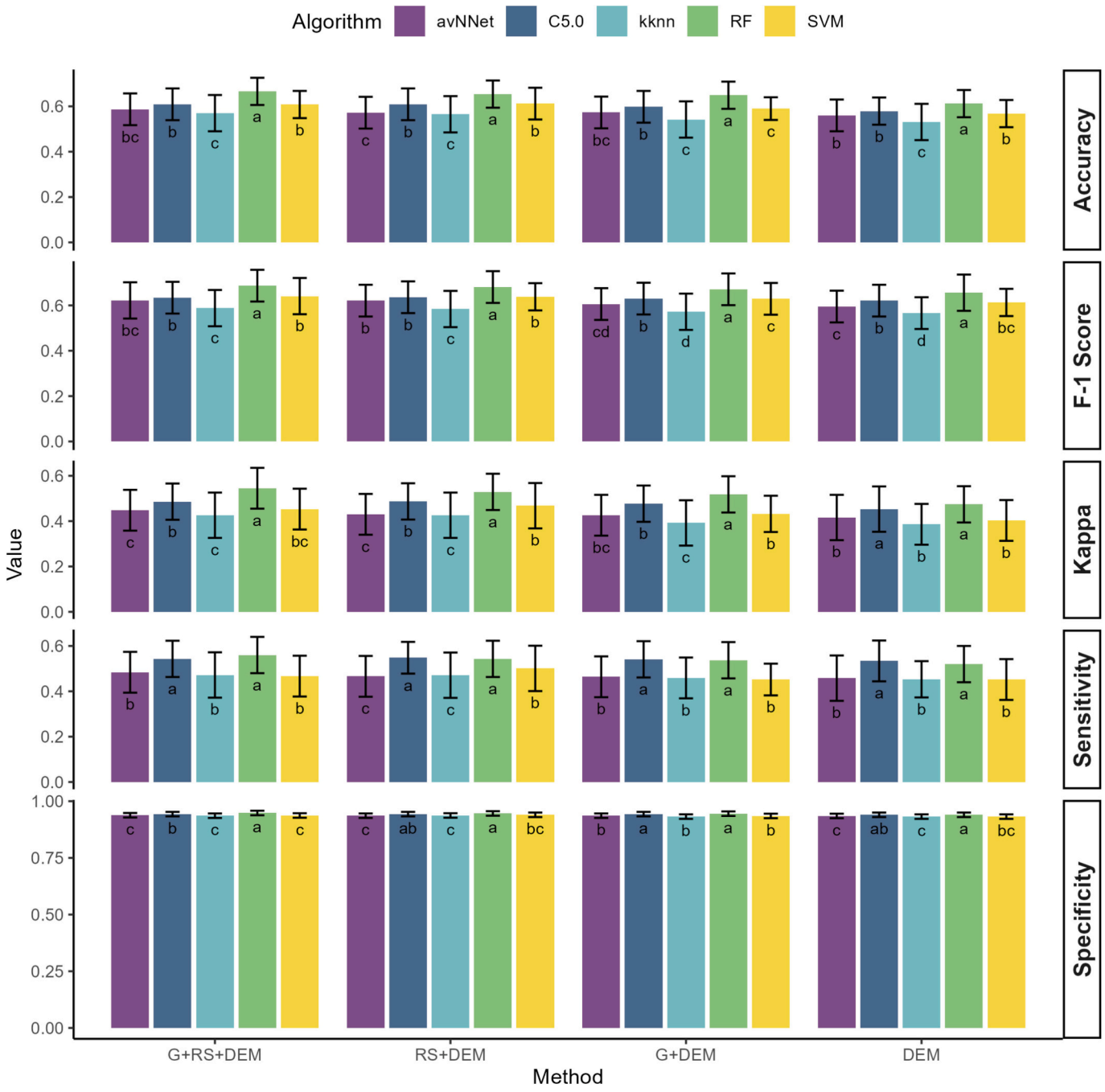


Fig. 4. Model's performance in terms of F1-Score, accuracy, kappa, sensitivity, and specificity for predicted digital pedological mapping.

with discrepancies were classified as errors. Then, the final percentage of errors and successes was computed. This process was carried out in R using the *terra* package.

3. Results

3.1. Model's performance and importance of variables

The Random Forest (RF) algorithm achieved the highest prediction performance for soil types and all strategies evaluated, compared to other tested algorithms, based on five evaluative parameters (Fig. 4 **and Appendix A**). On the other hand, the K-nearest neighbor (kknn) algorithm obtained the lowest prediction performance for soil classes (Fig. 4 **and Appendix A**). The other algorithms tested achieved intermediate performance in the predictive modeling of pedological classes (Fig. 4 **and Appendix A**). The Kruskal-Walli's test presented a significant statistical difference between the RF algorithm and the other models tested, as well as among the other models themselves (Fig. 4 **and Appendix A**).

The G + RS + DEM strategy for predictive pedological mapping using the RF algorithm demonstrated statistically superior performance compared to the all-other strategies across all metrics and showed either superiority or equivalence to the RS and G strategies in specific metrics, such as sensitivity and specificity (Table 3). Consequently, G + RS + DEM was chosen as the best strategy because it demonstrated the highest performance metrics and statistically significant differences compared to all other strategies. In addition, the performance metric values for the RS + DEM; G + DEM, and DEM strategy were statistically similar according to the Kruskal-Walli's test, despite the numerical differences in Accuracy, Kappa, and Specificity values (Table 3).

RF: Random Forest algorithm; G + RS + DEM (geophysical + remote sensing strategy); RS + DEM (remote sensing strategy; G + DEM (geophysical strategy); DEM (Digital elevation model strategy).

Fig. 5 shows, in detail, the relative percentage contribution of each covariate used in soil map prediction by ML algorithms, including geophysical variables, relief attributes, and remote sensing data. The environmental variables that contributed most to the modeling of pedological classes (>50% contribution) using the RF algorithm varied across the predictor combinations. For the G + SR + DEM and SR + DEM strategies, *relief* was the most influential variable, followed by spectral indices from the SYSI dataset (mainly GREEN and SWIR1 bands) and morphometric parameters such as *STANH* and *TWI*. In the G + DEM strategy, *relief* and *magnetic susceptibility* exhibited the highest importance, accompanied by *STANH* and *Th/K*, indicating a strong interaction between topography and gamma-spectrometric ratios. For the DEM-only strategy, *relief* again dominated the model, while *STANH*, *SolRation1*, *SolDiffuse1*, and *TWI* also showed substantial contributions, highlighting the role of terrain morphology, solar radiation dynamics, and hydrological indices in explaining the spatial variability of pedological classes (Fig. 5).

Table 3
Performance of the best Random Forest (RF) models using four different combinations of covariates (G + RS + DEM; RS + DEM; G + DEM; and DEM only), evaluated using the F1 score, accuracy, Kappa coefficient, sensitivity, and specificity, for the scenario without RS sensors.*

Parameter of performance	Algorithms			
	RF (G + RS + DEM)	RF (RS + DEM)	RF (G + DEM)	RF(DEM)
F-1 Score	0.687 ± 0.07 a	0.681 ± 0.07 ab	0.671 ± 0.07 ab	0.656 ± 0.08 ab
Accuracy	0.666 ± 0.06 a	0.654 ± 0.06 a	0.649 ± 0.06 a	0.612 ± 0.06 b
Kappa	0.545 ± 0.09 a	0.529 ± 0.08 a	0.518 ± 0.08 a	0.474 ± 0.08 b
Sensitivity	0.560 ± 0.08 a	0.543 ± 0.08 ab	0.537 ± 0.08 ab	0.520 ± 0.08 b
Specificity	0.948 ± 0.01 a	0.946 ± 0.01 a	0.945 ± 0.01 a	0.940 ± 0.01 b

* Different letters within the same row indicate values that do not differ significantly, as determined by the Kruskal-Wallis's test followed by Dunn's post hoc test at a 5% significance level.

3.2. Predicting soil classes and comparing with reference maps

The best strategy for pedological mapping was achieved using a combination of geophysical sensors and remote sensing techniques (G + RS + DEM) predicted by RF algorithm (Fig. 5). The maps in Fig. 5 demonstrate the best correspondence in the colored pedological units in the lower layer for each strategy using the RF model. The lines in the upper layer represent the actual pedological classes observed and delineated through traditional soil mapping by an experienced pedologist, with the G + RS + DEM strategy showing the highest correspondence (more details show Fig. 1 in supplementary material.). There was an 72.9% correspondence between the predicted soil types and the pedological classes observed in the base map, using the Agreement evaluation method of validation (Fig. 7, Table 4). The RS + DEM and G + DEM strategies for soil mapping each matched 69.66% of the predicted soil map compared to the base pedological map (Fig. 6). Comparing the isolated use of DEM in predictive modeling of soil types, the correspondence of the predicted pedological map with the reference soil map was 69.56%, based on the Agreement validation method (Fig. 7, Table 4). Finally, the smallest soil map prediction errors occurred with the G + RS + DEM (27.37%), followed + DEM, RS + DEM (30.34%), G + DEM strategies (29.29%), and the comparative DEM (30.44%), using the Agreement/Disagreement evaluation method (Fig. 7, Table 4).

Regarding the prediction error of pedological classes using the RF algorithm, the G + RS + DEM strategy achieved the highest accuracy, with approximately 54% of the mapped area showing errors below 5% and about 20% within the 5–20% range. The G + DEM configuration presented a similar performance, with roughly 53% of the area in the lowest error class and 20% showing moderate errors. The DEM-only model also maintained satisfactory accuracy, with about 50% of the area exhibiting minimal error and 22% moderate error levels. In contrast, the RS + DEM strategy showed comparatively lower accuracy, with 44.6% of the area in the lowest error class and a higher proportion of moderate to high errors (>20%). Overall, these results indicate that the integration of geophysical variables, particularly gamma spectrometry, with remote sensing and terrain attributes, enhances model robustness and spatial reliability, reducing uncertainty in soil class prediction across heterogeneous landscapes (Fig. 8).

The confusion matrix demonstrated a clear correspondence between the predicted and observed pedological classes. Among the evaluated strategies, the G + RS + DEM combination achieved the highest predictive performance, with an overall agreement of 0.729 and the lowest disagreement value (0.273). The G + DEM model showed an equivalent level of accuracy (overall agreement of 0.729), confirming the strong influence of geophysical variables on model stability. The RS + DEM strategy presented slightly lower agreement (0.698) and higher disagreement (0.302), while the DEM-only configuration exhibited the weakest performance, with an overall agreement of 0.697 and disagreement of 0.304. These results indicate that incorporating both geophysical and spectral data with terrain attributes enhances the RF model's capacity to delineate pedological classes more reliably, reducing classification uncertainty across complex geomorphological settings (Table 4, Fig. 9).

In all strategies used, the pedological classes that presented the largest prediction errors were Rhodic Acrisol (PV), Haplic Acrisol (PVA), Xanthic Acrisol (PA), and Lithic Leptosol (RL), which were erroneously classified as Haplic Cambisol (CX) (Fig. 9). Additionally, the pedological classes Rhodic Acrisol (PV) and Lithic Leptosol (RL) were misclassified as Rhodic Nitisol (NV) in all mapping strategies (Fig. 9). All mapping strategies showed greater prediction accuracy for the pedological classes



Fig. 5. Importance of predictor variables (geophysical data, remote sensing data, and relief attributes) for pedological predictive maps generated using the Random Forest algorithm. X-axis: Relative percentage of contribution, displaying only variables with contributions exceeding 50%. Y-axis: variables that contributed described in Table 2. The colors represent the groups to which each variable belongs. Variables shown in green correspond to morphometric parameters derived from the Digital Elevation Model (DEM), those in yellow are derived from remote sensing (RS) data, and those in purple represent geophysical (G) variables. Contribution represents the normalized contribution of each covariate to the model prediction, expressed as a percentage. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 4
Statistical "agreement accuracy" and "disagreement accuracy" for all strategies used.

Statistical performance parameter	RF (G + RS + DEM)	RF (RS + DEM)	RF (G + DEM)	RF (DEM)
Overall Agreement	0.729	0.698	0.729	0,697
Overall Disagreement	0.273	0.302	0.273	0,304

Haplic Cambisol (CX), Haplic Plinthosol (FX), Dystric Gleysol (GX), Rhodic Ferralsol (LV), and Rhodic Nitisol (NV), with these classes presenting the highest percentage accuracy values on the diagonal of the matrices (Fig. 9).

4. Discussion

The RF algorithm demonstrated superior performance in predictive

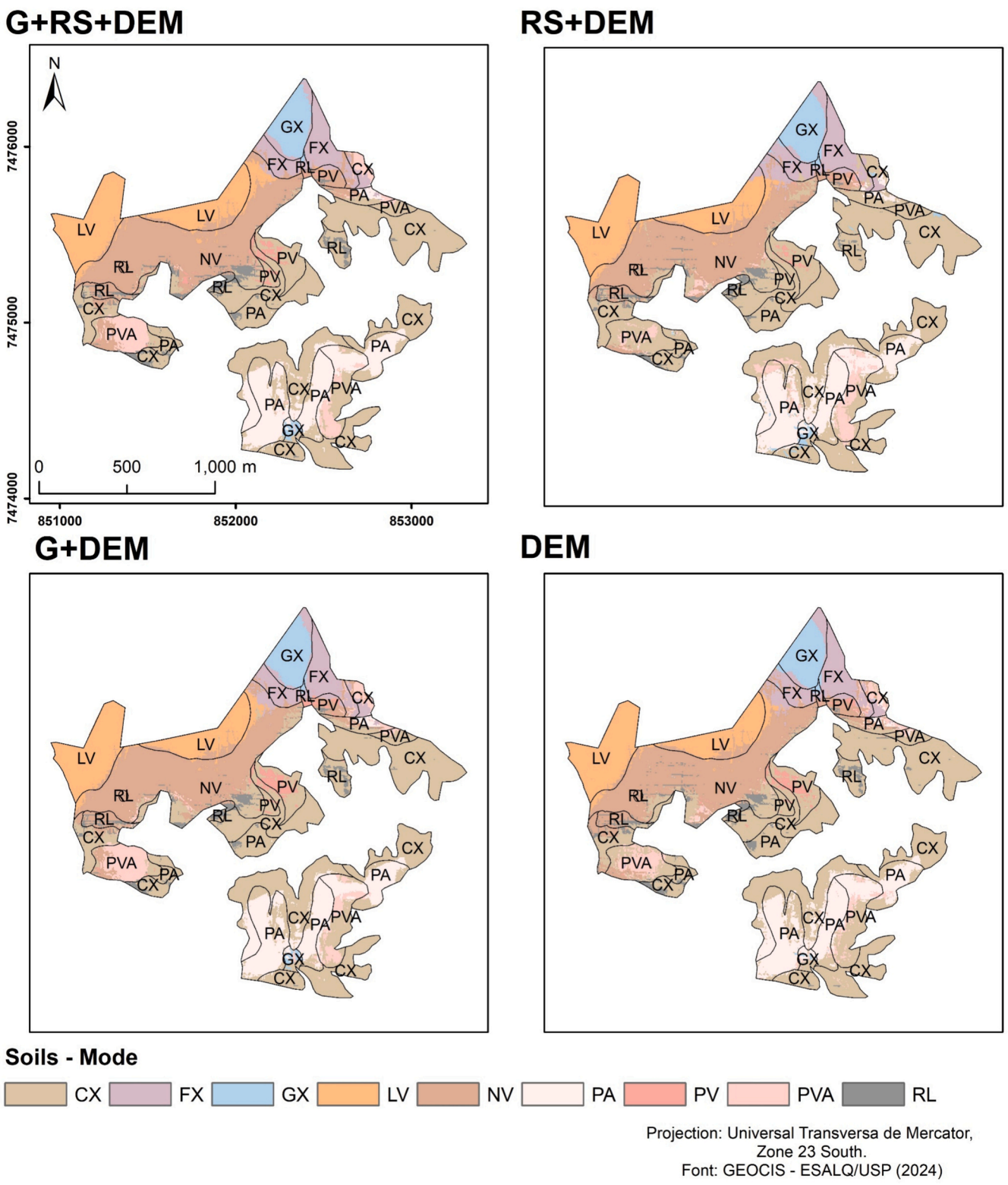


Fig. 6. Pedological maps predicted by the Random Forest (RF) model for each strategy used: G + RS (geophysics + remote sensing), RS (remote sensing), G (geophysics), and comparative DEM (digital elevation model - relief variables). The colors in the lower layer of the map indicate the pedological classes predicted for each strategy by the RF model. The lines in the upper layer indicate the actual pedological classes observed and delineated in traditional soil mapping by an experienced pedologist. CX: Haplic Cambisol; FX: Haplic Plinthosol; GX: Dystric Gleysol; LV: Rhodic Ferralsol; NV: Rhodic Nitisol; PA: Xanthic Acrisol; PV: Rhodic Acrisol; PVA: Haplic Acrisol; RL: Lithic Leptosol.

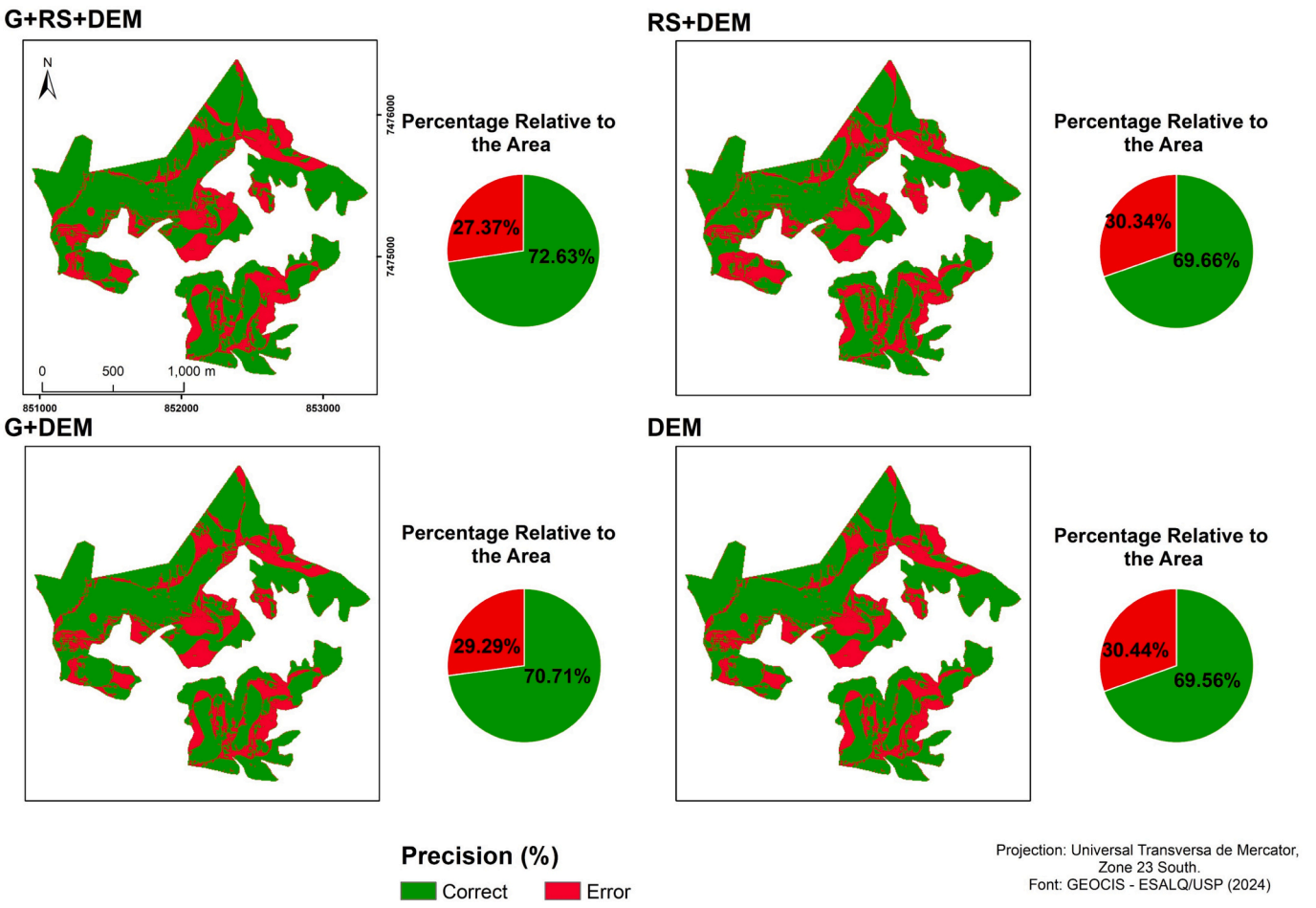


Fig. 7. Percentage of success and error based on the Agreement/Disagreement statistical validation test between the maps predicted by the Random Forest model and those observed through traditional pedological mapping, for each predictive pedological mapping strategy used G + RS (geophysics + remote sensing), RS (remote sensing), G (geophysics), and DEM (digital elevation model - relief variables).

modeling of pedological classes across all evaluated strategies, using geophysical data collected in the field and unprocessed relief data as input for the model. Additionally, the values of the model's performance metrics may have been affected by the heterogeneity of the area in terms of lithology, soils, and relief, as well as the low number of samples and their distribution within the area, which may not represent the spatial variations of soil-lithology-relief characteristics (Islam et al., 2022; Mello et al., 2022b). The RF algorithm is one of the decision trees algorithm families are considered satisfactory generalist methods, which achieve consistent results even on small datasets as in our case due to their robustness in the prediction (Wang et al., 2018). This results in excellent generalization capability, enabling the algorithm to minimize errors and enhance the precision of the predictive pedological model (Deka, 2014).

Our findings align with previous studies demonstrating that ensemble learning algorithms, such as Random Forest, are robust to multicollinearity and heterogeneous data sources, making them particularly suitable for multi-sensor data fusion (Bellinaso et al., 2021; Mello et al., 2023c; Mello et al., 2025Mello et al., 2022c; Pham et al., 2023; Vingiani et al., 2022). The superior performance of the G + RS + DEM strategy underscores the importance of combining datasets that capture complementary soil formation factors geophysical signals linked to parent material and subsurface structure, and spectral–morphometric variables associated with pedogenic processes and surface conditions. The relatively small performance differences observed between integrated and single-sensor models suggest that certain soil–landscape relationships are already well captured by relief and spectral variables, and that geophysical data add marginal gains mainly in areas with complex lithological variability.

Among the variables considered in the predictive models across all strategies, the most important were the relief, standardized height (STANH), SISY and soil κ. Relief, derived from the digital elevation model, represents the topographic configuration of the terrain surface and delineates landform features that directly influence surface runoff, drainage, and erosion–deposition dynamics. The STANH expresses the relative vertical position of a point along the slope, quantifying the distance between the local base level and the upper slope limit. This variable captures geomorphological gradients that affect soil development by controlling energy of relief, drainage, and redistribution of materials along the slope. The SYSI integrates multi-temporal satellite imagery to generate a composite spectral representation of bare soil surfaces, minimizing vegetation and moisture effects. By capturing the long-term spectral response of soils under varying environmental conditions, SYSI reflects differences in mineralogy, texture, organic matter, and moisture content, attributes closely linked to pedogenetic processes such as weathering, leaching, and oxidation–reduction (Demattê et al., 2018; de Sousa et al., 2024). Consequently, SYSI serves as a proxy for soil genesis and properties, enabling the identification of spatial variations associated with parent material composition, degree of weathering, and surface stability across the landscape. Soil κ is governed by the ferrimagnetic mineralogical composition of the soil, primarily controlled by the parent material's iron content and by redox dynamics along the relief (Cervi et al., 2019; Shenggao, 2000). Well-drained upland positions generally exhibit higher κ values due to the concentration

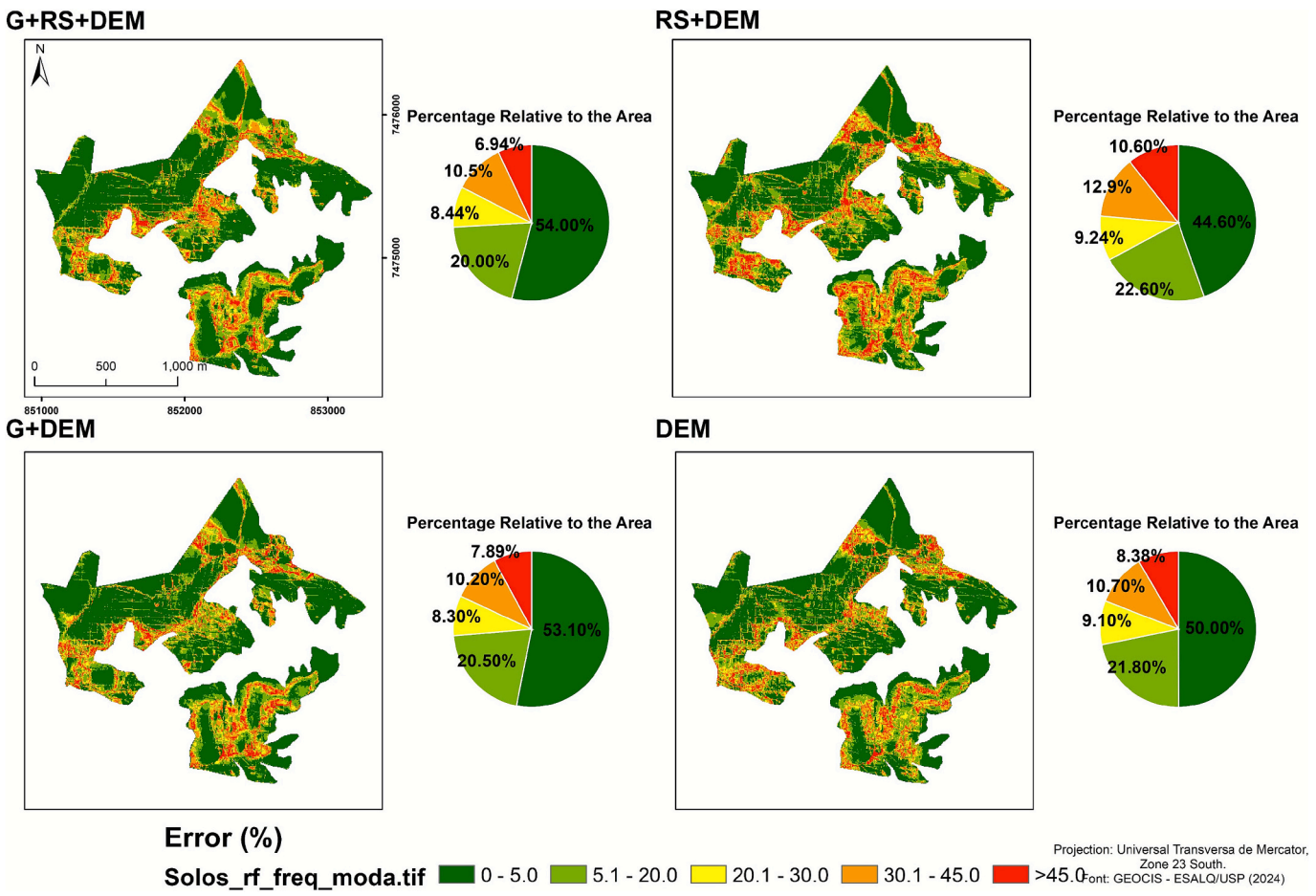


Fig. 8. Prediction error maps of pedological classes (number of times the algorithm did not predict the pedological modal class) generated by the Random Forest algorithm for each evaluated predictive pedological mapping strategy and the digital elevation model. G + RS (geophysics + remote sensing), RS (remote sensing), G (geophysics), and DEM (digital elevation model - relief variables).

or formation of ferrimagnetic minerals (e.g., magnetite, maghemite), whereas poorly drained or hydromorphic settings promote their transformation into weakly magnetic phases, leading to reduced κ values (Mello et al., 2020).

The best strategy for digital pedological mapping was the combined use of geophysical data and remote sensing (G + RS + DEM) as input data in the predictive modeling of pedological classes, using the RF algorithm. The combination of geophysical (G + DEM) techniques with remote sensing (RS + DEM) likely achieved better performance in soil mapping due to the complementarity of soil information. Geophysical techniques detect more subsurface pedological attributes, such as parent material, soil layers, and diagnostic horizons (Mello et al., 2023d; de Mello et al., 2022a, 2022b; Ramos et al., 2017; Ribeiro et al., 2018; Triantafilis et al., 2009; Wilford and Minty, 2006), while RS captures surface information, including variations in relief, soil color, and moisture (Demattê et al., 2018; Mello et al., 2023; Silvero et al., 2021). Our results suggest that, together, these techniques provide insights into both soil-forming factors and pedological processes, with geophysical variables highlighting subsurface differences and relief variables emphasizing surface variations. In addition, combining geophysical data with remote sensing information enhances the model's ability to capture complex patterns of soil types, leading to better predictive performance in pedological mapping. Combining G and RS data in the G + RS + DEM strategy maximizes the amount and diversity of information available to the model. In contrast, isolated strategies (G + DEM, RS + DEM, or DEM) provide more limited information, resulting in lower performance. The G + RS approach is effective because the geophysical variables, including radionuclides (eU, eTh, and K^{40}), soil κ, and soil ECa of the soil, are closely associated with lithology (mineralogical composition of the rocky substrate) and specific attributes of each soil type mapped in the area (Dickson and Scott, 1997; Heggemann et al., 2017; Jordanova, 2016; Loiseau et al., 2020; Lu et al., 2012; Sarmast et al., 2017; Sudduth et al., 2005). These attributes mainly include texture, textural relationships between horizons, mineralogy, base saturation, and the presence of ions.

The strategy using only geophysical variables (G) in predictive pedological modeling demonstrated similar performance to that achieved using remote sensing alone (RS + DEM), suggesting that integrated geophysical techniques (gamma spectrometry, magnetic susceptibility, and apparent electrical conductivity) contributed comparably to the RS strategy. Notably, RS is more closely associated with soil surface attributes (Fathololoumi et al., 2020; Taghdis et al., 2024) whereas geophysical techniques are more strongly linked to subsurface attributes (Mello et al., 2020; Mello et al., 2021; Mello et al., 2022c(de Mello et al., 2022c)). Detected geophysical variables are related to certain soil attributes, which in turn are connected to diagnostic horizons used for soil classification. Meanwhile, RS detects soil surface attributes that are indicative of subsurface attributes (Mello et al., 2021). Therefore, the combination of these mapping strategies tends to be more efficient, especially for geophysical variables.

The mapping strategy using only remote sensing (RS + DEM) presented no significant statistical difference compared to the geophysical strategy (G + DEM), which the performance evaluation metrics of the RF algorithm were numerically similar for the RS strategy. The SYSI images

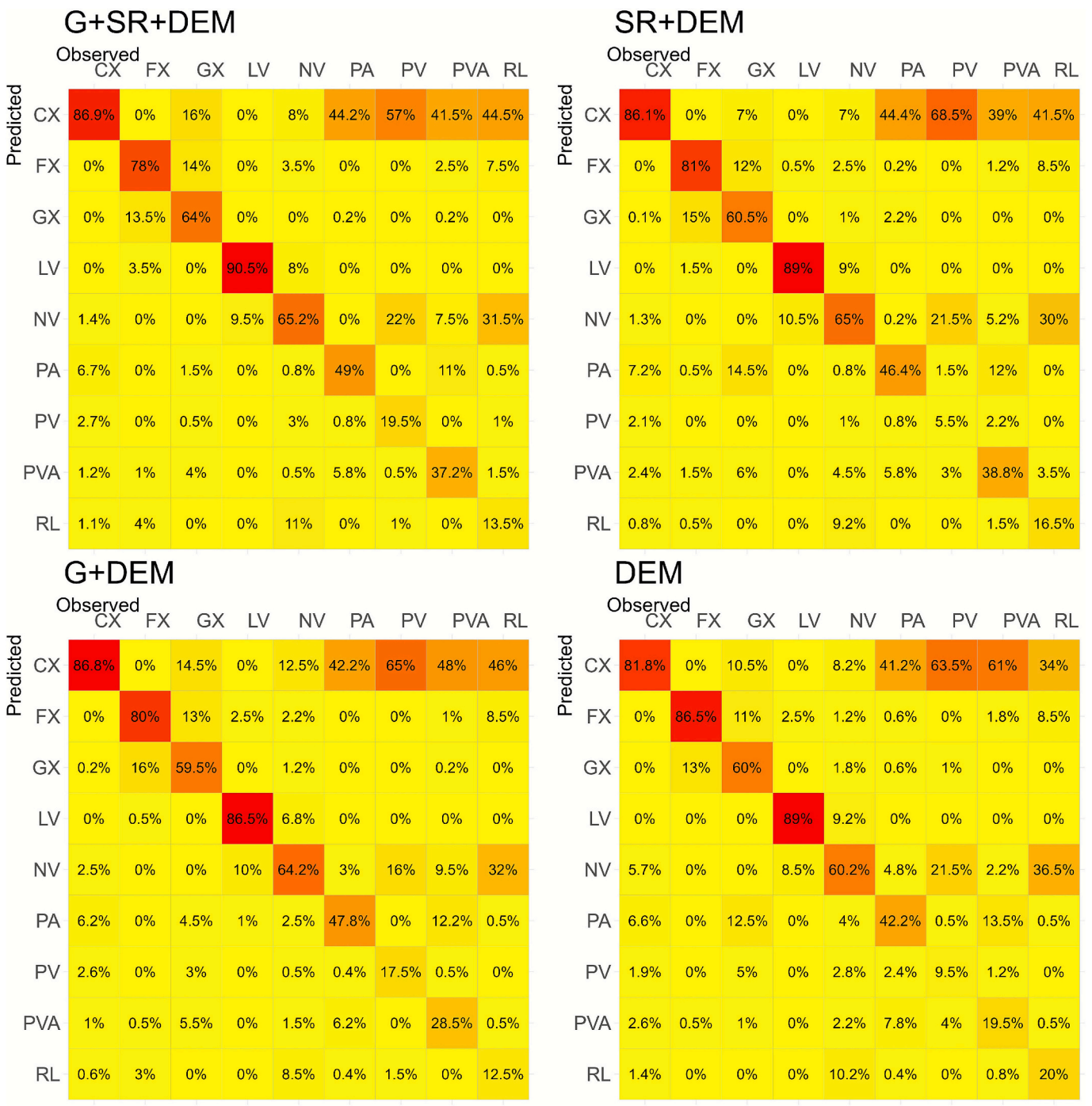


Fig. 9. Confusion matrix demonstrating the producer's accuracy in percentage, indicating the probability that an observed sample was correctly classified within the predicted pedological class. The columns indicate the percentage accuracy of the predicted samples. Proposed and evaluated predictive pedological mapping strategies and the digital elevation model. G + RS (geophysics + remote sensing), RS (remote sensing), G (geophysics), and DEM (digital elevation model - relief variables).

are correlated with soil surface reflectance, which is influenced by factors such as color, texture, oxic mineralogy, and moisture (Demattê et al., 2018). The spatial variability of soils within the study area, particularly in terms of color, texture, and moisture, was distinctly captured in the SYSI image. Additionally, variations in relief patterns and features significantly influence pedogenesis, leading to differences in topsoil composition. For instance, redder soils indicate a higher hematite content compared to yellow soils. These features provide insights that extend beyond the information typically captured by conventional digital elevation models (DEMs). This underscores the capability of SYSI to reveal additional details about the soil surface that are often overlooked in DEMs. Therefore, red (oxic) soils with a clayey texture, such as Rhodic Ferralsol, Rhodic Nitisol, Rhodic Acrisol, and Haplic Cambisol (abbreviated as LV, NV, PV, and CX, respectively), located on diabase and claystone, will exhibit a distinct spectral behavior compared to soils with a sandy texture located on sandstones or siltite such as Lithic Leptosol, Xanthic Acrisol, Haplic Acrisol, and Haplic Cambisol (RL, PA, PVA, and CX, respectively). The grayish soils of hydromorphic environments (GX) have a different mineralogy and texture than all the aforementioned soils and, consequently, a different spectral behavior

(Demattê et al., 2017). Thus, the pedological classes with different colors, mineralogy, and texture, where these morphological attributes were used for soil classification, were efficiently stratified in the predictive modeling by the tested mapping strategies.

The confusion matrix demonstrated that the prediction errors in pedological classes were associated with morphological attributes common to both the predicted and observed classes. For example, considering the best mapping strategy, G + RS, the model showed good performance when modeling the pedological class CX (84.3%) but presented errors with classes PA, PV, PVA (37.2%, 66%, and 34.2%, respectively), and RL (35%), which share a predominantly sandy texture, at least on the surface, with CX on sandstone. Similarly, the clay-textured NV was predicted with 74.5% accuracy, but errors were noted with the pedological classes PV and RL (17.5% and 25.5%, respectively). The model likely confused the color between NV and PV and the clay content between NV and RL, which are derived from Diabase. Additionally, the pedological classes CX, FX, GX, LV, and NV had a greater number of samples compared to the classes PA, PV, PVA, and RL. Also, the prediction error indicates instances where the algorithm does not select the correct modal pedological class (de Almeida et al., 2021; Meier et al., 2018). This affects predictive modeling, as the RF algorithm is susceptible to sample size, tending to prioritize pedological classes with a larger number of samples due to its generalist nature (Siqueira et al., 2022).

4.1. Limitations, uncurtains and applicability to other environments

Although our study provides valuable insights into different digital soil mapping strategies by combining remote sensing, geophysical data, and soil formation factors (relief and lithology), it does have some limitations. First, the covariates used as input data in the predictive pedological mapping model (geophysical data, relief attributes, and SYSI) may not fully capture the spatial and temporal variability of soil formation factors, processes, and mapping units. This limitation arises from the resolution and availability of field-collected data. Second, while the Random Forest model proved effective, it assumes independence between predictors and is susceptible to *overfitting*, particularly given the relatively small size of our training dataset. Third, the reference data used for model validation were derived from classical soil mapping methods, whose error rates were not quantified. Although the Agreement/Disagreement method offers appropriate statistical support for validation, this lack of error quantification may introduce biases and affect the accuracy of the model's predictions. To address these limitations, future studies should incorporate higher-resolution datasets, explore alternative classification algorithms, and implement robust uncertainty quantification techniques to enhance predictive performance and reliability.

The methodology presented in this study is highly reproducible and can be readily applied to other study sites with similar tools and conditions across different environments, provided geophysical variables, SYSI, and relief data are available. The step-by-step approach, utilizing easily accessible covariates such as remote sensing data and lithological maps, ensures broad applicability across diverse regions. Furthermore, the digital pedological mapping process, including soil sample collection and morphological profile descriptions, can be adapted to areas with comparable or different soil and climate conditions. The machine learning techniques and spatialization methods employed this study are transferable to other regions with relevant environmental data, offering valuable insights for similar applications. Additionally, the performance evaluation framework can serve as a benchmark for researchers seeking to assess the effectiveness of the methodology in different contexts.

Although the reference dataset was generated by an experienced soil expert, it still represents an interpretative classification rather than an absolute ground truth. Unlike quantifiable soil properties measured through laboratory analyses, soil class identification involves a degree of expert judgment, which may introduce subjective uncertainty into model validation. Nonetheless, this limitation is inherent to most digital soil mapping studies and remains acceptable within the scope of this research.

While geophysical sensors offer valuable insights into subsurface conditions, they entail substantially higher operational and logistical costs compared to freely available remote sensing data (e.g., Sentinel-2, Landsat, or SYSI). In our study, the marginal improvement in mapping accuracy (<5%) when incorporating geophysical data suggests that, for regional or exploratory surveys, the use of free satellite and morphometric data may be more cost-effective. Conversely, proximal geophysical surveys are justified in detailed site-specific investigations or calibration efforts where subsurface differentiation is critical (e.g., for agricultural zoning or geological boundary detection in pedogenetic studies). This highlights the need to balance data quality and economic feasibility, emphasizing that digital soil mapping frameworks should be scalable depending on project objectives, available resources, and required spatial resolution.

5. Conclusions

This study demonstrated that the most effective strategy for predictive pedological mapping is the integration of geophysical variables with remote sensing, lithology, and relief data (G + RS + DEM). The approaches using remote sensing, lithology, and relief data (RS), or geophysical data combined with lithology and relief (G + DEM), provided comparable performance in pedological mapping. However, the combination of geophysical data and remote sensing significantly enhanced the model's ability to identify complex soil type patterns, offering a richer dataset and improving predictive accuracy. In contrast, standalone approaches, such as those using only geophysical data (G), remote sensing (RS), or relief data (DEM), performed less effectively due to their narrower informational scope.

Among the five algorithms tested, the RF algorithm exhibited the highest performance metrics, suggesting its suitability for soil class prediction. Geophysical variables, primarily linked to subsurface soil diagnostic horizons, provided valuable insights into soil properties beneath the surface, while remote sensing data captured surface characteristics and pedogeomorphological features. Notably, environmental variables like relief, SYSI, and morphometric factors, and soil κ which govern water flow and soil moisture, emerged as key drivers of pedogenesis across all mapping strategies, though their relative importance varied.

The digital pedological maps generated by machine learning algorithms aligned significantly with traditional pedological maps, with accuracy ranging from 69.56% to 100%. The lowest accuracy was observed at 69.56%, with errors concentrated in areas with lithological-pedological transitions. Specific pedological classes (PA, PV, PVA, and RL) exhibited moderate modeling errors, often related to soil morphological attributes such as texture and color, which can be challenging to capture accurately via remote sensing.

Beyond methodological insights, these results have practical implications for resource allocation in soil surveys. Our analysis demonstrates that while multi-sensor integration slightly improves predictive accuracy, the cost–benefit ratio must be carefully considered, especially when low-cost and openly accessible data already achieve satisfactory mapping performance. Thus, this study contributes to establishing quantitative benchmarks for evaluating when geophysical data integration is warranted in future digital soil mapping initiatives.

In conclusion, integrating geophysical techniques with remote sensing holds considerable potential for improving large-scale soil mapping and surveying. Unlike traditional methods, which are limited to soil profiles, the G + RS + DEM strategy enables the analysis of both vertical and lateral soil variation, offering deeper insights into pedogenesis. Future studies and fieldwork can benefit from this integrated approach, enhancing the precision and scope of soil mapping efforts.

CRediT authorship contribution statement

Gustavo Vieira Veloso: Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Danilo César de Mello:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Lucas Vieira Silva:** Writing – review & editing, Visualization. **Elpídio Inácio Fernandes-Filho:** Writing – review & editing, Methodology. **Jorge Tadeu Fim Rosas:** Validation, Methodology, Data curation. **Fellipe Alcantara de Oliveira Mello:** Writing – review & editing, Investigation. **José João Lelis Leal de Souza:** Writing – review & editing, Investigation. **Márcio Rocha Francelino:** Writing – review & editing, Investigation. **Sara Ramos dos Santos:** Writing – review & editing, Formal analysis. **Francis Henrique Tenório Firmino:** Writing – review & editing, Investigation. **Nícolas Augusto Rosin:** Writing – review & editing, Investigation. **Gabriel Pimenta Barbosa de Sousa:** Writing – review & editing. **Tiago Osório Ferreira:** Writing – review & editing, Investigation. **Arnaldo Barros e Souza:** Writing – review & editing, Investigation. **José A.M. Demattê:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

Statement: During the preparation of this work, the authors utilized GPT-4 to correct the grammar and structure of the English language, ensuring clarity in the sentences for the reader. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.catena.2026.109822.

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